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CO1-ASSIGNMENT I - BASED ON SDG GOALS

Explain the challenges in representing multi-modal data (satellite imagery + weather data) as neural network inputs for wildfire prediction systems.

(SDG 15 – Life on Land)

Problem Understanding

Wildfires are becoming more frequent and severe due to climate change, rising temperatures, prolonged droughts, and changing weather conditions. Predicting wildfires accurately is difficult because it requires combining satellite imagery (land cover, vegetation, fire hotspots) with weather data (temperature, humidity, wind speed, and rainfall), which differ in format, resolution, and timing. These differences make it challenging for neural networks to process and learn from the data effectively. Therefore, there is a need for an efficient multi-modal neural network model that can integrate both satellite imagery and weather information to provide accurate and timely wildfire predictions, helping to protect forests, biodiversity, and ecosystems in support of SDG 15 – Life on Land.

Dataset

Dataset Name: NASA FIRMS + ERA5 Weather Dataset

Description:
The wildfire prediction model uses a combination of satellite imagery and weather data to predict the likelihood of wildfire occurrence.

Data Sources:

- NASA FIRMS (Fire Information for Resource Management System): Satellite images and active fire hotspot data.
- ERA5 Weather Dataset: Temperature, humidity, wind speed, rainfall, and atmospheric pressure.

Dataset Features:

Feature	Description
Latitude	Geographic location
Longitude	Geographic location
Land Cover	Type of vegetation/land
NDVI	Vegetation health index
Surface Temperature	Ground temperature (°C)
Humidity	Relative humidity (%)
Wind Speed	Wind speed (m/s)
Rainfall	Precipitation (mm)
Fire Radiative Power (FRP)	Fire intensity
Confidence	Confidence score of fire detection
Brightness	Thermal brightness from satellite
Target	0 = No Wildfire, 1 = Wildfire

Expected Output

- 0 → No Wildfire
- 1 → Wildfire

This dataset combines satellite imagery and weather parameters, making it suitable for training neural networks to predict wildfire risk while supporting SDG 15 – Life on Land.

Import Libraries

```
# ================================
# Import Libraries
# ================================
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
# ================================
# Load Dataset
# ================================
df = pd.read_csv("wildfire_dataset.csv")
# Display Dataset
print(df.head())
print(df.info())
print(df.describe())


# ================================
# Check Missing Values
```

```
# ==================================
print(df.isnull().sum())
# Fill Missing Values
df.fillna(df.median(numeric_only=True), inplace=True)
# ==================================
# Select Features and Target
# ==================================
X = df[['Temperature','Humidity','WindSpeed','Rainfall','NDVI']]
y = df['Fire']
# ==================================
# Feature Scaling
# ==================================
scaler = StandardScaler()
X = scaler.fit_transform(X)
# ==================================
# Split Dataset
# ==================================
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# ==================================
# Build Neural Network
```

```
# ================================
model = Sequential()
model.add(Dense(32, activation='relu', input_shape=(5,)))
model.add(Dense(16, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
# ================================
# Compile Model
# ================================
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
# ================================
# Train Model
# ================================
history = model.fit(
    X_train,
    y_train,
    epochs=20,
    batch_size=16,
    validation_split=0.2
)
#================================
# Prediction
```

```
# ================================
y_pred = model.predict(X_test)
y_pred = (y_pred > 0.5).astype(int)
# ================================
# Evaluation
# ================================
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report")
print(classification_report(y_test, y_pred))
print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))
# ================================
# Accuracy Graph
# ================================
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Wildfire Prediction Accuracy")
plt.legend()
plt.show()
# ================================
# Loss Graph
# ================================
plt.plot(history.history['loss'], label='Training Loss')
```

```
plt.plot(history.history['val_loss'], label='Validation Loss')

plt.xlabel("Epoch")

plt.ylabel("Loss")

plt.title("Wildfire Prediction Loss")

plt.legend()

plt.show()
```

Concept Learning

The neural network learns patterns from satellite imagery and weather data to identify conditions that may lead to wildfires. After training, it can classify new data as Wildfire (1) or No Wildfire (0) based on the learned patterns.

Split Dataset

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
```

Version Spaces and Candidate Elimination

Version space represents all hypotheses that are consistent with the wildfire training data. Candidate Elimination gradually removes inconsistent hypotheses until only the valid hypotheses remain for predicting wildfire occurrence.

1. Inductive Bias

   The learning algorithm assumes that the patterns learned from the training data will also apply to unseen satellite images and weather conditions for accurate wildfire prediction.

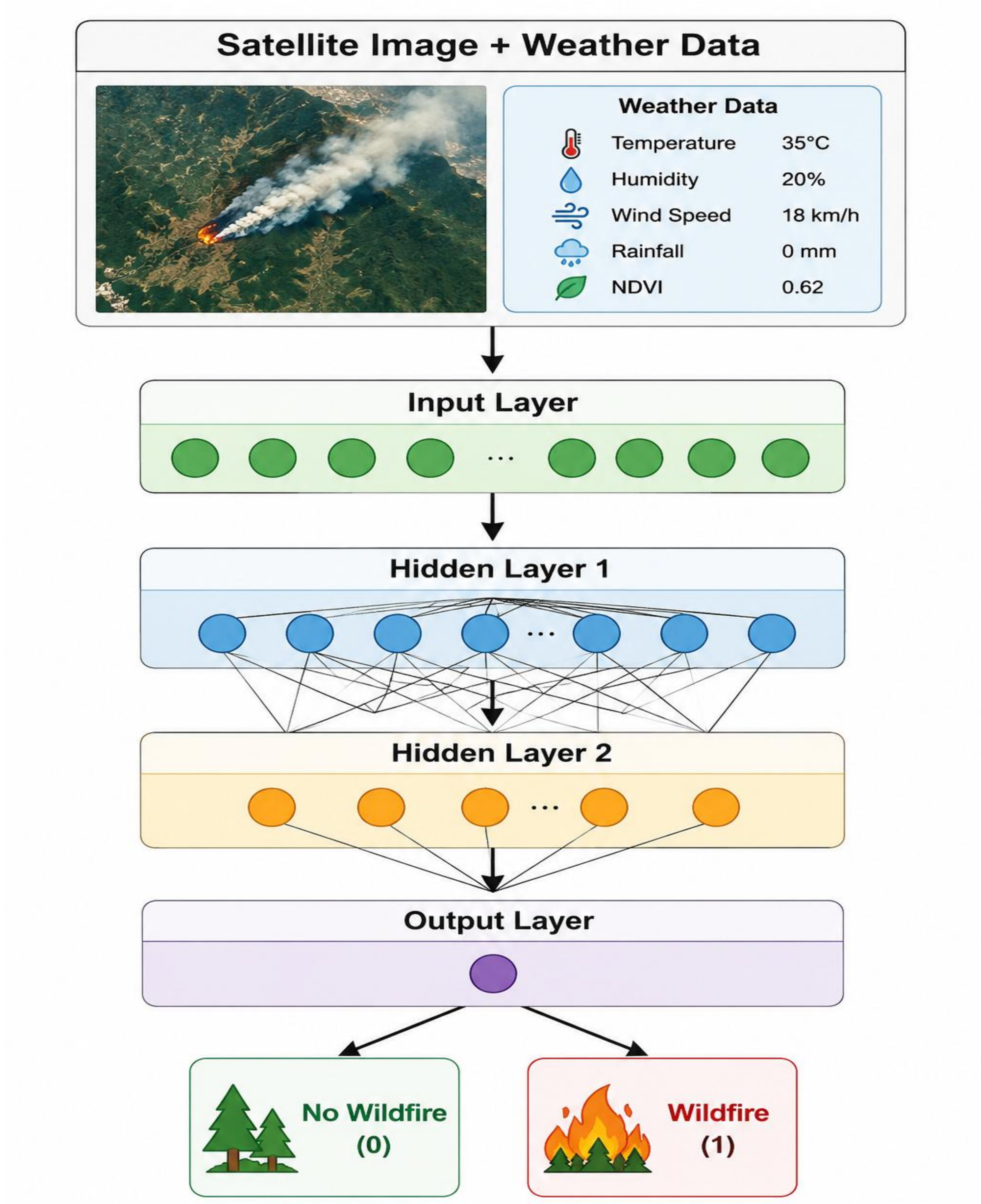
2. Neural Network Learning

   A Neural Network learns complex patterns by combining features extracted from satellite imagery and weather data. It automatically adjusts its weights during training to accurately classify wildfire and non-wildfire conditions.

3. Representation Algorithm

   The learned knowledge is represented as interconnected layers of neurons, where each layer extracts meaningful features and passes them to the next layer, enabling accurate wildfire prediction.

Neural Network Representation



Heuristic Space Search

The neural network uses optimization techniques such as Gradient Descent and the Adam optimizer to minimize prediction errors by updating network weights during training. This process helps the model learn the best patterns from satellite imagery and weather data for accurate wildfire prediction.

SDG Mapping

This project supports SDG 15 (Life on Land) by improving early wildfire prediction, protecting forests and biodiversity, reducing environmental damage, assisting disaster management authorities, and promoting sustainable ecosystem conservation.

This project supports SDG 15 by:

- Early detection of wildfires
- Protecting forests and wildlife
- Reducing environmental damage
- Assisting disaster management
- Promoting sustainable ecosystem conservation

Conclusion

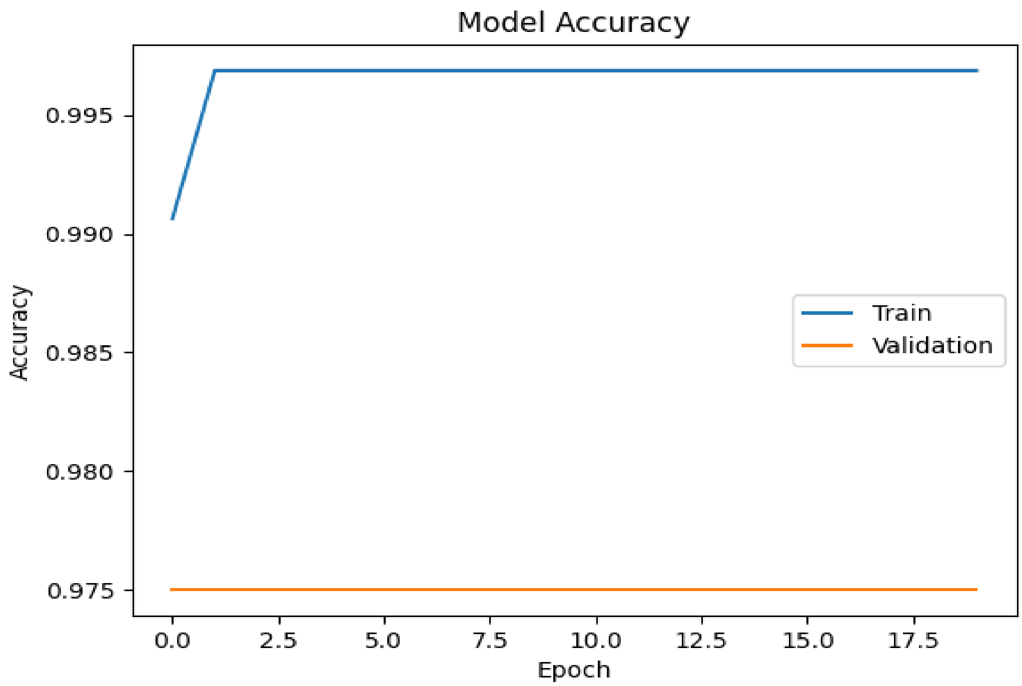
The wildfire prediction model demonstrates how a neural network can combine satellite imagery and weather data to accurately predict wildfire occurrences. Proper data preprocessing and model training improve prediction accuracy, helping in early wildfire detection, protecting forests, conserving biodiversity, and supporting SDG 15 – Life on Land.

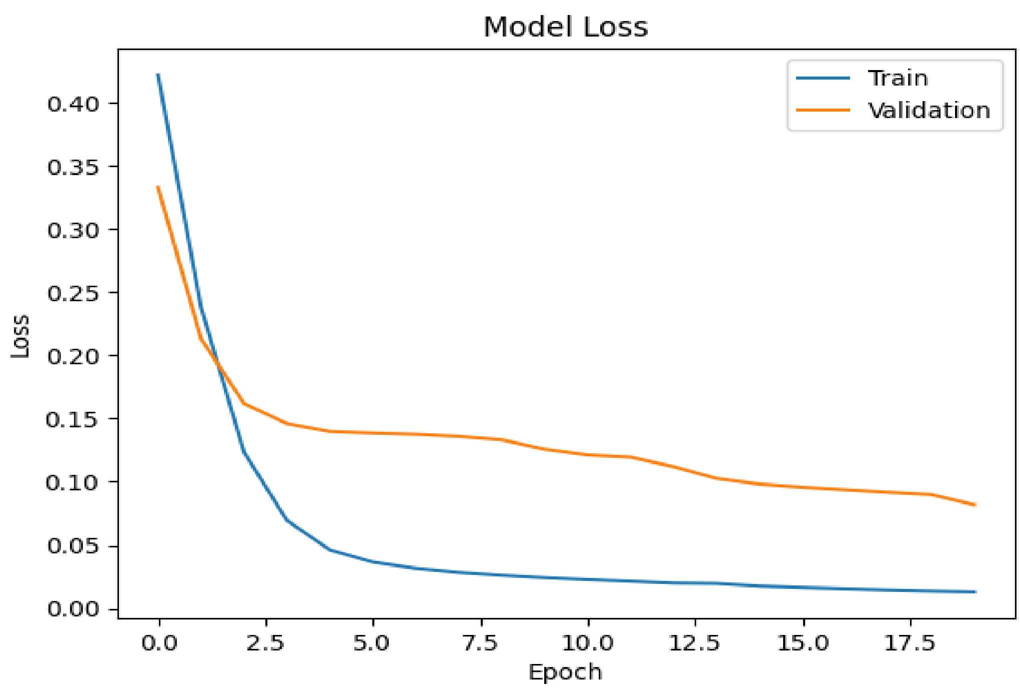
```
Classification Report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       100

    accuracy                           1.00       100
   macro avg       1.00      1.00      1.00       100
weighted avg       1.00      1.00      1.00       100
```

Visualization & Interpretation





```
   Temperature  Humidity  WindSpeed  Rainfall  NDVI  Fire
0           21        36         23         3  0.89     0
1           34        11         22        14  0.33     0
2           43        26          8        19  0.39     0
3           29        42         22        18  0.41     0
4           25        18          2         2  0.47     0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Temperature  500 non-null    int64
 1   Humidity     500 non-null    int64
 2   WindSpeed    500 non-null    int64
 3   Rainfall     500 non-null    int64
 4   NDVI         500 non-null    float64
 5   Fire         500 non-null    int64
dtypes: float64(1), int64(5)
memory usage: 23.6 KB
None
       Temperature    Humidity  WindSpeed    Rainfall       NDVI        Fire
count   500.000000  500.000000  500.00000  500.000000  500.00000  500.000000
mean     30.218000   48.670000   12.24200   10.050000    0.49032    0.006000
std       9.451409   23.166841    7.75498    5.880532    0.22307    0.077304
min      15.000000   10.000000    0.00000    0.000000    0.10000    0.000000
25%      22.000000   28.000000    5.00000    5.000000    0.30000    0.000000
50%      31.000000   48.000000   12.00000   10.000000    0.48500    0.000000
75%      39.000000   68.250000   19.00000   15.000000    0.66250    0.000000
max      45.000000   90.000000   25.00000   20.000000    0.90000    1.000000
Temperature    0
Humidity       0
WindSpeed      0
Rainfall       0
NDVI           0
Fire           0
```